

Day 5

☒ Wireless Networks Fundamentals

1. Wireless Network Types ☆☆☆

Type	Meaning	Range	Examples
WPAN	Wireless Personal Area Network	Very short	Bluetooth, Wi-Fi Direct
WLAN	Wireless Local Area Network	~100 m / few hundred feet	Wi-Fi / 802.11
WWAN	Wireless Wide Area Network	Miles / large geographic area	Cellular, satellite

Remember:

P → Personal → short

L → Local → building/campus

W → Wide → city/country



2. IEEE Wireless Standards

Standard	Frequency	Important fact
802.11a	5 GHz	54 Mbps
802.11b	2.4 GHz	11 Mbps

Standard	Frequency	Important fact
802.11g	2.4 GHz	54 Mbps
802.11n	2.4/5 GHz	MIMO, up to 600 Mbps
802.11ac	5 GHz	High-speed Wi-Fi
802.11ax	2.4/5 GHz	Modern Wi-Fi 6
802.11ad	60 GHz	Very high speed, short range

Must remember

- 802.11 = Wi-Fi
- 802.15 = WPAN/Bluetooth
- 802.16 = WiMAX



3. Frequency Bands ☆☆☆

2.4 GHz

- Longer range
- Better wall penetration
- More interference
- Fewer non-overlapping channels
- Common channels: **1, 6, 11**

5 GHz

- Higher bandwidth
- Less congestion
- More channels
- Shorter range
- Poorer wall penetration than 2.4 GHz

60 GHz

- Very high speed
- Very short range
- Used by **802.11ad**



4. RF Characteristics

Wireless communication uses **radio frequency (RF)**.

Important characteristics:

- RF is **unguided/unbounded**
- Signals can be affected by:
 - Walls
 - Distance
 - Weather
 - Electronic devices

- Other wireless networks
- Signal strength decreases with distance.
- **Multipath fading** occurs when signals take different paths and interfere with each other.



5. WLAN vs Ethernet

WLAN	Ethernet
Wireless medium	Wired medium
Uses CSMA/CA	Traditionally uses CSMA/CD
Cannot easily detect collisions while transmitting	Collision detection was used in shared Ethernet
802.11	802.3
Radio waves	Electrical/optical signals

VERY IMPORTANT

Wi-Fi → CSMA/CA → Collision Avoidance

Old shared Ethernet → CSMA/CD → Collision Detection



6. Wireless Network Architectures

Home Wireless Router

Usually combines:

1. Router
2. Switch
3. Access Point (AP)

So one device can provide:

Wi-Fi + LAN switching + routing/default gateway



Autonomous AP

- Standalone AP
- Configured individually
- Good for small networks
- No centralized controller required



Controller-Based Architecture

Uses:

- **Lightweight AP (LAP)**

- Wireless LAN Controller (WLC)
- CAPWAP

The **WLC** provides centralized management.

Remember:

LAP → WLC → Centralized control



Cloud-Managed AP

Example: **Cisco Meraki**

- Central management through cloud dashboard
- Easy configuration and monitoring
- Useful for large deployments



7. Wireless Antennas

Omnidirectional

□ Sends signal approximately **360°**

Good for:

- Offices
- Classrooms
- Open areas

Directional

□ Focuses signal in a particular direction.

Good for:

- Long-distance links
- Specific buildings
- Outdoor connections

Yagi

A type of **directional antenna**.



8. MIMO

MIMO = Multiple-Input Multiple-Output

Uses multiple antennas to transmit and receive multiple data streams.

Example:

- $1 \times 1 \rightarrow 1$ transmit / 1 receive
- $2 \times 2 \rightarrow 2$ transmit / 2 receive

- 3×3 → 3 transmit / 3 receive
- 4×4 → 4 transmit / 4 receive

Main benefit:

➔ Higher throughput and better wireless performance



9. Wireless Topologies

Ad Hoc / IBSS

Devices communicate **directly**.

PC ↔ Laptop

Copy

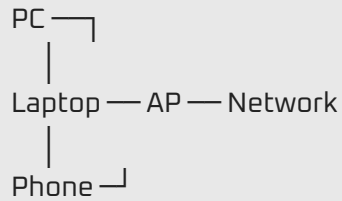
No AP required.

IBSS = Independent Basic Service Set



Infrastructure Mode

Clients communicate through an **Access Point**.



This is the most common WLAN architecture.



10. BSS, BSA, BSSID

These are **very likely exam questions**.

BSS

Basic Service Set

A single AP and its associated wireless clients.

BSA

Basic Service Area

The physical area covered by the AP's wireless signal.

BSSID

Basic Service Set Identifier

Usually the **MAC address of the AP's wireless radio**.

Memorize:



BSS = network group

BSA = coverage area

BSSID = AP MAC address



11. ESS

ESS = Extended Service Set

An ESS combines **two or more BSSs** connected through a **Distribution System (DS)**.

Example:

Copy

```
graph TD
    BSS1[BSS 1] --- AP1[AP1]
    AP1 --- DS[DS]
    DS --- AP2[AP2]
    AP2 --- BSS2[BSS 2]
```

Both APs can provide the **same SSID**, allowing clients to roam between them.

Key terms

- **BSS** → one AP
- **ESS** → multiple BSSs
- **SSID** → network name
- **BSSID** → AP radio MAC address
- **DS** → connects BSSs



12. SSID vs BSSID ☆☆☆

SSID

➔ **Wireless network name**

Example:

AAiT-WiFi

BSSID

➔ **MAC address identifying a particular BSS/AP radio**

Easy memory:

SSID = Name

BSSID = MAC



13. CSMA/CA

Wi-Fi uses:



Carrier Sense Multiple Access with Collision Avoidance

Basic idea:

1. Listen to the channel.
2. If busy → wait.
3. If available → transmit according to contention rules.
4. ACK confirms successful reception.

RTS/CTS

RTS = Request To Send

CTS = Clear To Send

Client — RTS → AP
Client ← CTS — AP
Client — DATA → AP



RTS/CTS helps reduce problems such as the **hidden-node problem**.

Exam trap ⚠

RTS/CTS is **not required for every Wi-Fi transmission**. It can be used when configured/needed.



14. AP Discovery

There are two main methods.

Passive Scanning

AP sends **Beacon frames**.

AP → Beacon → Clients

Copy

The client listens.

Remember:

Passive = AP announces



Active Scanning

Client sends a **Probe Request**.

Client → Probe Request → AP
AP → Probe Response → Client

Copy

Remember:

Active = Client asks



15. Authentication & Association

The basic connection process:

1. Discovery

Find an AP.

2. Authentication

Client authenticates.

3. Association

Client associates with the AP.

Memorize:



Discover → Authenticate → Associate



16. Wireless Channels

2.4 GHz

The commonly used non-overlapping channels are:



1, 6, 11

Memory trick:

1 → 6 → 11

These are separated enough to avoid overlap under common 20/22 MHz channel configurations.



17. Channel Bonding

Channel bonding combines channels to increase bandwidth.

802.11n

Can use:

20 MHz → 40 MHz

802.11ac

Can use:

- 20 MHz
- 40 MHz
- 80 MHz
- 160 MHz

Main benefit:

➔ **Higher throughput**

Important trade-off:

Wider channels can consume more spectrum and may reduce the number of independent channels available.



18. Spread Spectrum Technologies

OFSSS

Direct-Sequence Spread Spectrum

- Spreads data over a wider frequency band.
- Used by older 802.11b.
- Helps with interference resistance.



FHSS

Frequency-Hopping Spread Spectrum

- Rapidly changes frequencies.
- Used historically by wireless technologies including Bluetooth.

Memory:

FHSS = Frequency Hopping



OFDM

Orthogonal Frequency-Division Multiplexing

- Divides a channel into multiple orthogonal subcarriers.
- Improves spectral efficiency.

Used in many Wi-Fi standards including:



19. Wireless Security

Security Evolution

Remember the order:

Open → **WEP** → **WPA** → **WPA2** → **WPA3**

Open

× No meaningful wireless encryption/authentication protection.

WEP

- × Very insecure
- × Uses RC4
- × Easily compromised
- × Should not be used

WPA

Uses:

TKIP

Better than WEP but now obsolete.

WPA2

Uses:

AES/CCMP

Much stronger.

WPA3

Modern Wi-Fi security with improved authentication/security protections.



20. Personal vs Enterprise Security

WPA-Personal

Uses:



PSK = Pre-Shared Key

Good for:

- Homes
- Small offices

Example:



WPA-Enterprise

Uses:

- **802.1X**
- **EAP**
- **RADIUS**

Good for:

- Universities
- Companies
- Large organizations

Memorize:

Personal → PSK

Enterprise → 802.1X + EAP + RADIUS



21. Wireless Security Threats

Rogue AP

An unauthorized AP connected to the organization network.

⚠ Can create a major security risk.



Evil Twin ☆☆☆☆☆

An attacker creates a fake AP using the **same or similar SSID** as a legitimate network.

Example:

Legitimate:

AAU-WIFI

Attacker:

AAU-WIFI

Users may connect to the fake AP.

Possible goal:

- ➡ Credential theft
- ➡ Traffic interception
- ➡ Man-in-the-Middle attack



Eavesdropping

Attacker listens to wireless traffic.

Protection:

➡ Strong encryption



DoS

Goal:



Make the wireless network unavailable.

Examples:

- RF interference/jamming
- Deauthentication attacks
- CTS flooding



22. Deauthentication Attack

Attacker sends **spoofed deauthentication frames**.

Result:

➔ Client gets disconnected from the AP.

Mitigation:

Management Frame Protection (802.11w / MFP) where supported.



23. CTS Flood

Attacker sends many fake:



CTS = Clear To Send

frames with long duration values.

Result:

➔ Other clients believe the medium is reserved and stop transmitting.



24. Weak Security Techniques

Hidden SSID

× Not real security.

MAC filtering

× Can be bypassed through MAC spoofing.

WEP

× Broken/insecure.

Best approach:

➔ Use modern WPA2/WPA3 security and proper authentication.



25. Wireless Intrusion Prevention

WIPS

Wireless Intrusion Prevention System

Can help detect/prevent:

- Rogue APs
- Wireless attacks
- Unauthorized devices
- Suspicious wireless activity





26. AP Placement

Good AP placement:

- ☐ High position
- ☐ Near ceiling
- ☐ Away from obstructions
- ☐ Consider user density
- ☐ Avoid major interference sources

High-density area

Place APs **where users actually are**, such as conference rooms/classrooms, rather than simply placing them in hallways.



27. Coverage Overlap

For roaming between APs:

➡ Adjacent coverage areas should overlap.

Your notes emphasize approximately:

10–15% overlap

This helps clients maintain connectivity while moving between APs.





28. Common Interference Sources

Examples:

- Microwave ovens
- Cordless phones
- Baby monitors
- Other APs
- Physical obstacles
- Competing RF devices

Important:

2.4 GHz is generally more crowded than 5 GHz.



29. Troubleshooting Wireless Networks

Use a structured approach.

Bottom-Up

Start from:

Physical → Data Link → Network → Transport → Application

Good when you suspect a physical/connectivity issue.



Top-Down

Start from:

Application → Transport → Network → Data Link → Physical

Useful when applications/services are failing.



Divide-and-Conquer

Start somewhere in the middle and test.

Example:

Client

↓

Can it reach the AP?

↓

Can it ping default gateway?

↓

Can it reach Internet?

↓

Copy

Is DNS working?



30. Important Troubleshooting Commands

Windows

```
ipconfig  
ipconfig /all  
ping
```

Copy

Linux

```
ip addr  
ip route  
ping
```

Copy

Check:

- IP address
- Subnet mask/prefix
- Default gateway

- DNS
- Wireless connection
- Security settings



31. Dual-Band Wi-Fi

2.4 GHz

Better for:

- Longer range
- Basic browsing
- Devices farther from AP

5 GHz

Better for:

- Higher throughput
- Streaming
- Less congestion
- Modern devices

Easy exam answer:

Need range → 2.4 GHz

Need speed/less congestion → 5 GHz

